

Discriminative Mixture Variational Autoencoder for Semisupervised Classification

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Abstract—In this article, a deep probability model, called the discriminative mixture variational autoencoder (DMVAE), is developed for the feature extraction in semisupervised learning. The DMVAE consists of three parts: 1) the encoding; 2) decoding; and 3) classification modules. In the encoding module, the encoder projects the observation to the latent space, and then the latent representation is fed to the decoding part, which depicts the generative process from the hidden variable to data. In particular, the decoding module in our DMVAE partitions the observed dataset into some clusters via multiple decoders whose number is automatically determined via the Dirichlet process (DP) and learns a probability distribution for each cluster. Compared to the standard variational autoencoder (VAE) describing all data with a single probability function, the DMVAE has the capacity to give a more accurate description for observations, thus improving the characterization ability of the extracted features, especially for the data with complex distribution. Moreover, to obtain a discriminative latent space, the class labels of labeled data are introduced to restrict the feature learning via a softmax classifier, with which the minimum entropy of the predicted labels for the features from unlabeled data can also be guaranteed. Finally, the joint optimization of the marginal likelihood, label, and entropy constraints makes the DMVAE have higher classification confidence for unlabeled data while accurately classifying the labeled data, ultimately leading to better performance. Experiments on several benchmark datasets and the measured radar echo dataset show the advantages of our DMVAE-based semisupervised classification over other related methods.

Index Terms—Dirichlet process (DP) mixture, discriminative representation, semisupervised learning, variational autoencoder (VAE).

I. INTRODUCTION

AS A DATA mining technology, statistical modeling with the probabilistic latent variable model has attracted extensive attention from the community of machine learning and pattern recognition. Probabilistic latent variable models learn the embedding with low dimension by creating a generative process from the latent space to the observation

space, thus capable of revealing intrinsic information of data. Du *et al.* [1] utilized the probability principal component analysis (PPCA) model to describe the statistical correlations among a different range cells of a 1-D radar high-resolution range profile (HRRP). Tunc *et al.* [2] proposed the class-dependent factor analysis (CDFA) by combining the factor analysis (FA) model with manifold learning for performing face recognition and Guo *et al.* [3] exploited the multiscale convolutional FA merged with the Bayesian support vector machine (SVM) to achieve the classification of images. In [4], a probabilistic topic model, that is, latent Dirichlet allocation (LDA), is proposed and applied to the text categorization.

However, the above probabilistic latent variable models are of shallow structures and fail to learn the deep representations, which have been proved to be more advantageous in various real-world tasks [5]–[7], [29]–[32]. In the combination of the merits of the deep network and probabilistic model, the variational autoencoder (VAE) [5] is proposed and used in many applications. For example, in the radar target recognition community, Zhai *et al.* [38] proposed a robust VAE (RVAE), whose recognition accuracy on three airplane targets has improved remarkably, such as approximately 3% higher than that of the shallow FA model. Based on the RVAE, a factorized discriminative conditional VAE (FDCVAE) is proposed by Du *et al.* [39], which also exploits the sample labels to constrain the learning of deep features, thus strengthening the discriminant power of the latent space and further improving recognition performance. In addition, in the field of anomaly detection, An and Cho [6] determined whether a sample is from the outlier by using the reconstruction error from the VAE. Since the deep feature extracted by the VAE can better reflect the nature of data, the greater reconstruction error in the VAE framework can be obtained when a sample comes from the outlier, compared with the shallow models. For example, the area under the curve (AUC) of the receiver operating characteristic (ROC) via the proposed method on the KDD-CUP99 dataset is 12% higher than that of the principal component analysis (PCA). Moreover, for the task of anomaly detection in multivariate time-series data, a combined version of the VAE and echo-state network, called echo-state VAE (ESVAE), is proposed in [40], which also outperforms the shallow methods.

Although the VAE makes the deep probabilistic model possible, one drawback of this method is that a single decoder has the limited capacity to accurately capture the distribution description of observed data and, thus, may lead to poor

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features, especially for a large-scale dataset whose distribution is often complicated. In data analysis, the mixture of expert (ME) [8] is selected as an alternative way to deal with the complex distributed data. In the ME architecture, the input space is split by a gate into several regions and a separate model, that is, expert, is learned for each region. This approach allows each expert to specialize in a smaller part of a larger problem, and multiple experts thus support the adequate description of complex data. Lima *et al.* [9] combined the ME with SVM to handle the nonlinear classification. Li [10] used ME to make a new type of hidden Markov model that the current state depends both on the preceding state and preceding observation. In [11], ME is also combined with the Gaussian process (GP) to deal with the problem of cumbersome matrix inversion.

By means of ME, we propose a mixture VAE to obtain the deep features that can more accurately represent observed data. The mixture VAE divides the dataset into some clusters with the generation of data in each cluster being described by a decoder. For an observed sample, which cluster it belongs to is dependent on the probability measures produced by all decoders. Since a single decoder can fit observations with simple probability distribution well, the collection of decoders offers the potential to make a more precise estimation for complex distributed data, which avoids the weak representation of deep features caused by the inaccurate depiction of observed data. In our method, the clustering is implemented via the Dirichlet process (DP) [11], which also allows for the self-adaptive determination of the cluster number.

All of the above VAE-based models are unsupervised and only focus on the description of observations without considering the supervised information. Actually, labels of a certain number of samples are often available in the classification task. To take full advantage of label information, we incorporate labels of labeled data into the mixture VAE and propose a discriminative mixture VAE (DMVAE). Besides the distribution depiction of labeled and unlabeled data via the collection of decoders, the DMVAE projects the latent features of labeled data to their labels by leveraging a classifier. The diversity of different class labels guides the learning of features in a more separable direction, which strengthens the discriminability of feature space. Moreover, for the unlabeled data, we also minimize the entropy of their predicted labels to enhance the generalization of the classifier, in the case of small labeled sample size.

Actually, similar to our DMVAE model, some work combining the clustering idea with VAE is also developed, and these studies can be divided into two classes: 1) the mixture of Gaussian-based VAE (MoG-VAE) [7], [33]–[35] and 2) the DP-based VAE (DP-VAE) [36], [37].

The MoG-VAE model replaces the Gaussian distribution in the traditional VAE with the Gaussian mixture distribution, which can better disentangle the representations from different classes. Following this line, Eric *et al.* [33] proposed a deep latent Gaussian mixture model (DLGMM), which uses the mixture of Gaussian distributions as the approximate posterior of the latent feature. Besides the feature's posterior, the

prior of the latent feature is also set to the Gaussian mixture distribution in the Gaussian mixture VAE (GMVAE) [7], with the prior distribution parameters in each component being learned via a network. In [34], an improved GMVAE, called variational deep embedding (VaDE), is developed by Jiang *et al.* Compared with the GMVAE, the VaDE not only captures the relationship between the component indicator and latent feature but also mitigates the information loss induced by the mean-field approximation, thus further enhancing the characterization capability of the latent feature. Based on the VaDE model, Yang *et al.* [35] proposed a deep clustering method via a Gaussian mixture VAE with graph embedding (DGG), which minimizes the Jensen–Shannon (JS) distances of features in each mixture component and obtains the more separable embeddings than the VaDE. Although these methods utilize more flexible latent feature distribution, that is, the mixture of Gaussian, to better represent the observed data, they only focus on the description of observations. When the data from different classes are highly similar, it is still difficult for the MoG-VAEs to obtain the separable features. However, our DMVAE also makes full use of the label information to further enhance the interclass separability of features while better describing the observed data, thus having the potential to learn a more discriminative latent space than the MoG-VAE models. Moreover, different from MoG-VAEs where the number of clusters is fixed as that of the classes, our DMVAE can automatically determine the cluster number according to data itself, which avoids the performance decrease in feature separability caused by the inappropriate cluster number setting. In addition, the MoG-VAEs are the unsupervised feature extraction methods and cannot directly use the classification task. Nevertheless, the DMVAE projects the latent feature to the class label with a softmax classifier, which realizes the joint learning of the latent feature and classifier.

For the DP-VAE method, the clustering process is implemented via the DP. Nalisnick and Smyth [36] proposed a Bayesian nonparametric version of the VAE, that is, stick-breaking VAE (SBVAE). SBVAE allows for the automatic determination of the feature dimension by imposing a Griffiths, Engen, and McCloskey (GEM) prior on the feature. In [37], an infinite VAE (IVAE) is proposed and applied to the semisupervised classification. The IVAE model uses the mixture of multiple independent VAEs to capture the complexity of observations, with the mixing coefficients being modeled by the DP distribution. The IVAE is first learned with the unlabeled samples, and then, can be used to determine the cluster membership of each labeled sample. Afterward, a classifier for each cluster is trained with the labeled samples belonging to the corresponding cluster, thus ultimately achieving the semisupervised classification. However, compared with our DMVAE, the IVAE has three major drawbacks.

- 1) The collection of multiple VAEs leads to the significant increase of model parameters.
- 2) In semisupervised learning, the IVAE may suffer the problem that the classifiers in some clusters cannot be trained when there are no labeled samples being assigned to the corresponding clusters. In this case,

the test samples divided into these clusters cannot be classified.

- 3) The clustering is independent of the classifier learning, which cannot guarantee that the data partition via the IVAE is most suitable for the training of the classifier.

The research reported here mainly focuses on the construction of the deep probability model, especially for the complex distributed data. Enlightened by the VAE and ME strategy, we develop a DMVAE that utilizes multiple decoders to describe the distribution of the whole dataset, which provides an opportunity for the latent features to express data more precisely. To learn a discriminative feature space, the label information of labeled data is introduced to the DMVAE via a classifier, with which the minimum entropy of predicted labels for unlabeled data can be guaranteed. The interplay among the data likelihood, label, and entropy constraints allows the DMVAE not only to accurately classify the labeled data but also to have higher classification confidence for unlabeled data, thus propitious to the improvement in generalization ability.

Table I lists all notations in our DMVAE model to assist readers clearly understand the matrix dimensions and definitions of notations.

The remainder of this article is organized as follows. In Section II, we give a brief review of the traditional VAE and DP. Then, the proposed DMVAE is introduced in detail in Section III. In Section IV, experiments on several benchmark datasets and measured radar dataset are conducted to validate the performance of our DMVAE. Finally, conclusions are summarized in Section V.

II. BACKGROUND

A. Variational Autoencoder

Given the dataset $\mathbf{X}$ consisting of N samples, representing as $\mathbf{X} = \{\mathbf{x}_n\}_{n=1}^N$, the VAE defines a generative scheme from the latent variable $\mathbf{z}_n$ to $\mathbf{x}_n$ via a decoder with parameter θ , that is, $p_\theta(\mathbf{x}_n|\mathbf{z}_n)$. Usually, the decoder is a deep neural network (NN) and outputs the distribution parameters of $p_\theta(\mathbf{x}_n|\mathbf{z}_n)$. The estimation of θ can be derived by maximizing the lower bound $\sum_{n=1}^N \text{LB}(\theta, \varphi; \mathbf{x}_n)$ of the log marginal likelihood $\ln p_\theta(\mathbf{x}_1, \dots, \mathbf{x}_N)$.

The lower bound $\text{LB}(\theta, \varphi; \mathbf{x}_n)$ can be expressed as

$$\text{LB}(\theta, \varphi; \mathbf{x}_n) = \mathbb{E}_{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)} [\ln p_\theta(\mathbf{x}_n|\mathbf{z}_n)] - \text{KL}(q_\varphi(\mathbf{z}_n|\mathbf{x}_n) \| p_\theta(\mathbf{z}_n)) \quad (1)$$

where the first term of right-hand side is the expectation of log likelihood $\ln p_\theta(\mathbf{x}_n|\mathbf{z}_n)$ with respect to (w.r.t.) the variational distribution $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$, $p_\theta(\mathbf{z}_n)$ is the prior distribution of the latent variable $\mathbf{z}_n$, and $\text{KL}(\cdot\|\cdot)$ denotes the Kullback–Leibler (KL) divergence. To deal with the problem of nonanalytical solution of the lower bound, caused by the complex form of $\mathbb{E}_{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)} [\ln p_\theta(\mathbf{x}_n|\mathbf{z}_n)]$, the VAE utilizes an encoder with parameter φ to represent $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$. The encoder takes $\mathbf{x}_n$ as the input and outputs the distribution parameters of $\mathbf{z}_n$ conditioned on $\mathbf{x}_n$. Therefore, the Monte Carlo estimator

TABLE I
LIST OF NOTATIONS IN THE DMVAE MODEL

Notation	Meaning	Notation	Meaning
$\mathbf{X}$	The observation dataset	c_n	The component indicator of $\mathbf{x}_n$ with $c_n \in \{1, \dots, K\}$
N	The number of samples	$\mathbf{c}_{-n}$	Indicator collection of all samples except $\mathbf{x}_n$
$\mathbf{x}_n$	The n th sample from $\mathbf{X}$	η_k^{-n}	The number of samples assigned to the k th component excluding $\mathbf{x}_n$
L	The number of labeled samples in $\mathbf{X}$	θ_k	Parameter of the k th decoder with $\boldsymbol{\theta} = \{\theta_k\}_{k=1}^K$
$\mathbf{x}_l$	The l th labeled sample	φ	The parameter of the encoder
$\mathbf{y}_l$	The class label of $\mathbf{x}_l$ with $\mathbf{Y} = \{\mathbf{y}_l\}_{l=1}^L$	ψ	The parameter of the softmax
U	The number of unlabeled samples in $\mathbf{X}$	$\boldsymbol{\mu}_{\theta_c}(\mathbf{z}_n)$	The mean of $\mathbf{x}_n$ conditioned on $\mathbf{z}_n$
$\mathbf{x}_u$	The u th unlabeled sample	$\boldsymbol{\sigma}_{\theta_c}(\mathbf{z}_n)$	The standard deviation of $\mathbf{x}_n$ conditioned on $\mathbf{z}_n$
$\mathbf{y}_u$	Predicted label of $\mathbf{x}_u$ with $\tilde{\mathbf{Y}} = \{\mathbf{y}_u\}_{u=1}^U$	$\boldsymbol{\mu}_\varphi(\mathbf{x}_n)$	The mean of $\mathbf{z}_n$ conditioned on $\mathbf{x}_n$
$\mathbf{z}_n$	The latent feature of $\mathbf{x}_n$	$\boldsymbol{\sigma}_\varphi(\mathbf{x}_n)$	The standard deviation of $\mathbf{z}_n$ conditioned on $\mathbf{x}_n$
J	The dimension of $\mathbf{z}_n$	$\lambda(\mathbf{z}_l)$	The output of the softmax with $\mathbf{z}_l$ as the input
I	The sampling number of $\mathbf{z}_n$	α	The hyper-parameter in DP
S	The number of classes	$\boldsymbol{\pi}$	The mixing coefficient with $\boldsymbol{\pi} = [\pi_1, \dots, \pi_K]$
$\boldsymbol{\varepsilon}_n$	The noise variable	β_1	The weight of label constraint
K	The number of decoders	β_2	The weight of entropy constraint

of the lower bound in (1) can be expressed as

$$\text{LB}(\theta, \varphi; \mathbf{x}_n) \simeq \frac{1}{I} \sum_{i=1}^I \ln p_\theta(\mathbf{x}_n|\mathbf{z}_n^i) - \text{KL}(q_\varphi(\mathbf{z}_n|\mathbf{x}_n) \| p_\theta(\mathbf{z}_n)) \quad (2)$$

where $\mathbf{z}_n^i$ is the i th sampling value of $\mathbf{z}_n$ sampled from $q_\varphi(\mathbf{z}_n|\mathbf{x}_n)$. With this approach, the lower bound has the explicit expression and can be optimized by performing stochastic gradient ascent (SGA).

However, the sampling of latent variable $\mathbf{z}_n$ hinders the error backpropagation, which makes the implementation of SGA impossible. To deal with this problem, a reparameterization trick is adopted, which introduces an auxiliary variable $\boldsymbol{\varepsilon}_n$ with the prior distribution $p(\boldsymbol{\varepsilon}_n)$ and represents $\mathbf{z}_n$ as

$$\mathbf{z}_n = g_\varphi(\mathbf{x}_n, \boldsymbol{\varepsilon}_n), \boldsymbol{\varepsilon}_n \sim p(\boldsymbol{\varepsilon}_n). \quad (3)$$

For example, taking Gaussian distribution as the posterior distribution of $\mathbf{z}_n$ with its mean $\boldsymbol{\mu}_\varphi(\mathbf{x}_n)$ and standard deviation $\boldsymbol{\sigma}_\varphi(\mathbf{x}_n)$ being encoded by the encoder, then we have

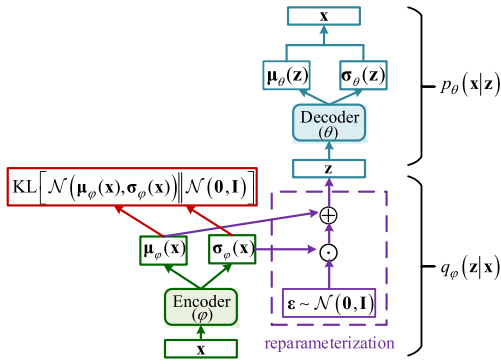


Fig. 1. Architecture of the VAE with Gaussian distribution assumption.

$\mathbf{z}_n = \mu_\phi(\mathbf{x}_n) + \sigma_\phi(\mathbf{x}_n) \odot \boldsymbol{\varepsilon}_n$ with $p(\boldsymbol{\varepsilon}_n) = \mathcal{N}(\boldsymbol{\varepsilon}_n | \mathbf{0}, \mathbf{I})$, $\mathcal{N}(\cdot)$ denoting the Gaussian distribution, and $\odot$ being the elementwise product operator. The sampling of $\mathbf{z}_n$ can thus be expressed as $\mathbf{z}_n^i = \mu_\phi(\mathbf{x}_n) + \sigma_\phi(\mathbf{x}_n) \odot \boldsymbol{\varepsilon}_n^i$, where $\boldsymbol{\varepsilon}_n^i$ is the i th sampling value of $\boldsymbol{\varepsilon}_n$. The parameterization technique avoids the direct sampling of $\mathbf{z}_n$, thus allowing backpropagation through the latent variable. For more intuitive explanation, we take the Gaussian distribution and standard normal prior as an example and present the structure of VAE in Fig. 1.

B. Dirichlet Process

DP [12] is a stochastic process that often applies to nonparametric Bayesian models. Let G_0 be a probability distribution in the measure space Φ and α be a positive parameter. For an arbitrary partition of Φ , represented as $A_1, A_2, \dots, A_r$, if there is a probability function G that brings the following relation into existence:

$$(G(A_1), \dots, G(A_r)) \sim \text{Dir}(\alpha G_0(A_1), \dots, \alpha G_0(A_r)) \quad (4)$$

with $\text{Dir}(\cdot)$ denoting the Dirichlet distribution, we say that G follows the DP composed of the baseline distribution G_0 and concentration parameter α , which can be expressed as $G \sim \text{DP}(G_0, \alpha)$.

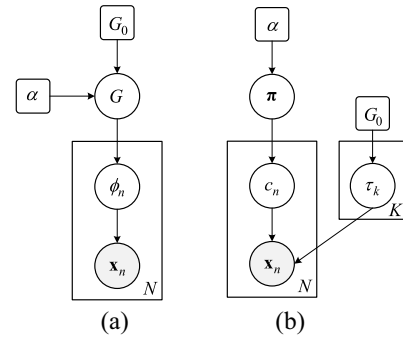
From the above definition, we know that the DP is a distribution placed over a distribution. Therefore, the sampling process based on DP consists of two phases: 1) drawing a sample distribution G from the DP and 2) pulling the samples $\{\phi_n\}_{n=1}^N$ from G , mathematically represented as

$$G|G_0, \alpha \sim \text{DP}(G_0, \alpha), \phi_n|G \sim G, n = 1, \dots, N. \quad (5)$$

Specifically, suppose that there are K different values $\{\tau_k\}_{k=1}^K$ for the first $N-1$ samples $\{\phi_n\}_{n=1}^{N-1}$, the probability distribution of the N th sample ϕ_N conditioned on $\{\phi_n\}_{n=1}^{N-1}$, G_0 , and α is

$$p(\phi_N | \{\phi_n\}_{n=1}^{N-1}, G_0, \alpha) = \sum_{k=1}^K \frac{l_k}{N-1+\alpha} \delta_{\tau_k} + \frac{\alpha}{N-1+\alpha} G_0 \quad (6)$$

where l_k denotes the number of samples with their values equal to τ_k in $\{\phi_n\}_{n=1}^{N-1}$, and δ_{τ_k} is the Kronecker delta function concentrated at τ_k (i.e., $\delta_{\tau_k} = 1$ if the value of ϕ_N is τ_k ; otherwise,

Fig. 2. Graphical representations of the (a) DPM model and (b) finite mixture model. Notations in the corner of the black solid boxes, including N and K , represent the boxes' stack for indices n and k .

$\delta_{\tau_k} = 0$). Equation (6) demonstrates that there is a certain probability for ϕ_N to take the same value as the previous samples, revealing the clustering property of the DP.

DP can cluster the data with the same values; however, it fails to perform clustering for those similar but different data, which strongly limits the application of DP. To deal with this problem, the DP mixture (DPM) model [13]–[21] is introduced. In the DPM model, the DP is used as the prior distribution of the observations, and for observed sample x_n , the DPM model can be expressed as

$$\mathbf{x}_n | \phi_n \sim p(\phi_n), \phi_n | G \sim G, G | G_0, \alpha \sim \text{DP}(G_0, \alpha) \quad (7)$$

where $\mathbf{x}_n$ is parameterized with ϕ_n . A graphical representation of the DPM model is shown in Fig. 2(a). There are three major approaches for the DPM model construction: one based on a Chinese restaurant process (CRP) model [15], [16], [19], one based on the stick-breaking conformation [12], [17], and one based on the infinite limit of finite mixture models [20], [21]. These three ways to represent DPM models are mathematically equivalent but have different formulations since they present the DPM from different perspectives, and in this article, we mainly focus on the infinite limit of finite mixture models.

Suppose a finite mixture model has K components and the mixing coefficient is $\pi = (\pi_1, \dots, \pi_K)$ with the Dirichlet prior $\text{Dir}(\alpha/K, \dots, \alpha/K)$. Let τ_k denote the parameter associated with mixture component k ($k = 1, \dots, K$) and its prior distribution is G_0 . For the purpose of drawing an observed sample $\mathbf{x}_n$ from the mixture model, it is essential to determine the component that $\mathbf{x}_n$ belongs to according to the mixing coefficient π . Let c_n denote the component indicator corresponding to $\mathbf{x}_n$, and then the finite mixture model can be represented as

$$\mathbf{x}_n | c_n, \{\tau_k\}_{k=1}^K \sim p(\tau_{c_n}), \tau_k | G_0 \sim G_0 \\ c_n | \pi \sim \text{Discrete}(\pi_1, \dots, \pi_K), \pi \sim \text{Dir}(\alpha/K, \dots, \alpha/K) \quad (8)$$

where $\text{Discrete}(\cdot)$ denotes the distribution of discrete variables. In Fig. 2(b), we give the directed-graph model of the finite mixture model. Ishwaran and Zarepos [20] have proven that the predicted distribution of indicator c_n is consistent with the property of DP and the finite mixture model is equivalent to the DPM model as $K \rightarrow \infty$, thus offering a way to approximately solve DP.

III. METHOD

A. Mixture Variational Autoencoder

The original VAE seeks to describe the observation by defining a generative process from the latent space to the data space via a decoder. Nevertheless, it is insufficient to capture the complexity of observed data well only with a single decoder, especially for the large-scale datasets. In this case, we refer to the idea of ME to obtain the most accurate description of observations with data clustering.

In this section, we propose a mixture VAE model. The mixture VAE estimates the probability distribution of observed data via multiple decoders, which divide the whole dataset into some clusters with each cluster being represented via a separate decoder. An ensemble of decoders has the capability to describe data more exquisitely, thus avoiding the weak separability of features caused by the single decoder's sweeping depiction. In addition, the number of clusters has the great effect on the model performance, and the improper setting of cluster number will lead to overfitting or underfitting. Considering the automatic determination of the cluster number, we adopt the infinite limit of finite mixture models-based DP to achieve the clustering task.

In detail, we posit that there are K ($K \rightarrow \infty$) components (decoders) for our mixture VAE with the k th component being parameterized by θ_k and $\theta = \{\theta_k\}_{k=1}^K$. Let c_n represent the component indicator of observed sample $\mathbf{x}_n$ and $\mathbf{c} = \{c_1, \dots, c_N\}$. By imposing a multinomial prior on $\mathbf{c}$, that is, $p(c_n) = \text{Mult}(\pi)$ with $\pi = (\pi_1, \dots, \pi_K)$, and a Dirichlet super-prior on π , that is, $p(\pi) = \text{Dir}(\alpha/K, \dots, \alpha/K)$ with α being the preset hyperparameter, the goal of the mixture VAE is to maximize the following expression:

$$\begin{aligned} p(\mathbf{X}, \mathbf{c}, \theta | \alpha) &= \iint p(\mathbf{X}, \theta | \mathbf{Z}, \mathbf{c}) p(\mathbf{Z}) p(\mathbf{c} | \pi) p(\pi | \alpha) d\pi d\mathbf{Z} \\ &= \int \left(\left[\prod_{n=1}^N p_{\theta_{c_n}}(\mathbf{x}_n | \mathbf{z}_n) p(\mathbf{z}_n) \right] \right. \\ &\quad \left. \times \left[\int p(\mathbf{c} | \pi) p(\pi | \alpha) d\pi \right] \right) d\mathbf{z}_n. \end{aligned} \quad (9)$$

Here, $\mathbf{Z} = \{\mathbf{z}_n\}_{n=1}^N$ and θ_{c_n} denotes the parameter of the decoder, to which $\mathbf{x}_n$ is assigned according to its indicator c_n . To render the proposed model tractable to variational inference, we optimize (9) by iteratively performing the following two steps.

1) *Optimize Joint Likelihood of $\mathbf{X}$ and θ Given the Current Indicator $\mathbf{c}$, That Is, $p(\mathbf{X}, \theta | \mathbf{c})$:* In this step, we adopt the solving trick used in the traditional VAE for finding the estimation of θ . Similar to the VAE, we have

$$\begin{aligned} \ln p(\mathbf{X}, \theta | \mathbf{c}) &= \sum_{n=1}^N \ln \left(\int p_{\theta_{c_n}}(\mathbf{x}_n | \mathbf{z}_n) p_{\theta_{c_n}}(\mathbf{z}_n) d\mathbf{z}_n \right) \\ &\geq \sum_{n=1}^N \mathbb{E}_{q_{\varphi}(\mathbf{z}_n | \mathbf{x}_n)} [\ln p_{\theta_{c_n}}(\mathbf{x}_n | \mathbf{z}_n)] \\ &\quad - \text{KL}(q_{\varphi}(\mathbf{z}_n | \mathbf{x}_n) \| p_{\theta_{c_n}}(\mathbf{z}_n)) \\ &= \text{LB}(\theta, \varphi; \mathbf{X}, \mathbf{c}). \end{aligned} \quad (10)$$

Equation (10) indicates that each observed sample $\mathbf{x}_n$ ($n = 1, \dots, N$) is generated from the mapping of the latent variable $\mathbf{z}_n$ via a specified decoder with parameter θ_{c_n} , and the posteriors of all latent variables $\{\mathbf{z}_n\}_{n=1, \dots, N}$ are approximated via an encoder parameterized by φ . Since the data distribution corresponding to each decoder is simpler than that of the whole dataset, each decoder in the mixture VAE has the ability to make a full description to its corresponding observations. Therefore, compared with the single probability function $p_{\theta}(\mathbf{X})$ in the traditional VAE, the joint probability density function [i.e., $\prod_{k=1}^K p_{\theta_k}(\mathbf{X}_k)$ with $\mathbf{X}_k$ representing the set of samples assigned to the decoder k] fitted by the collection of multiple decoders can more accurately represent the distribution of the whole dataset. As a result, the mixture VAE mitigates the loss of observations' details caused by the rough description of data and ultimately improves the quality of the features. As for the encoding network, its design is flexible since it is mainly used for the solution of intractable $\ln p(\mathbf{X}, \theta | \mathbf{c})$ and theoretically irrelevant to the motivation of our work, that is, the improvement of data description. Here, the reasons that we only use a single encoder rather than multiple encoders like the decoding network mainly comes from two aspects: 1) the single encoder can significantly reduce the number of parameters to be estimated and 2) in the semisupervised classification, the modeling approach of multiple encoders, that is, the observed data in each cluster are separately described by a VAE, may suffer the problem that the classifiers of some submodels cannot be trained when there are no labeled samples being assigned to the corresponding clusters. In this case, the test samples divided into these clusters cannot be classified.

Usually, $p_{\theta_{c_n}}(\mathbf{x}_n | \mathbf{z}_n)$ is set to a multivariate Gaussian distribution (in case of continuous data) or a Bernoulli distribution (in case of discrete data). Taking the Gaussian distribution as an example, the decoder for $\mathbf{x}_n$ outputs the mean and standard deviation conditioned on $\mathbf{z}_n$, that is, $\mu_{\theta_{c_n}}(\mathbf{z}_n)$ and $\sigma_{\theta_{c_n}}(\mathbf{z}_n)$. Then, $p_{\theta_{c_n}}(\mathbf{x}_n | \mathbf{z}_n)$ can be represented as

$$p_{\theta_{c_n}}(\mathbf{x}_n | \mathbf{z}_n) = \mathcal{N}(\mathbf{x}_n | \mu_{\theta_{c_n}}(\mathbf{z}_n), \text{diag}(\sigma_{\theta_{c_n}}^2(\mathbf{z}_n))) \quad (11)$$

where $\text{diag}(\cdot)$ denotes the diagonalization. Analogously, the approximate posterior $q_{\varphi}(\mathbf{z}_n | \mathbf{x}_n)$ can be achieved via the encoder, which has the form of

$$q_{\varphi}(\mathbf{z}_n | \mathbf{x}_n) = \mathcal{N}(\mathbf{z}_n | \mu_{\varphi}(\mathbf{x}_n), \text{diag}(\sigma_{\varphi}^2(\mathbf{x}_n))). \quad (12)$$

Furthermore, with the assumption of standard Gaussian prior $p_{\theta_{c_n}}(\mathbf{z}_n) = \mathcal{N}(\mathbf{z}_n | \mathbf{0}, \mathbf{I})$ ($n = 1, \dots, K$), the KL divergence can be explicitly expressed as

$$\begin{aligned} \text{KL}[q_{\varphi}(\mathbf{z}_n | \mathbf{x}_n) \| p(\mathbf{z}_n)] &= -\frac{1}{2} \sum_{j=1}^J \left[1 + \ln(\sigma_{\varphi}^{(n,j)})^2 - (\mu_{\varphi}^{(n,j)})^2 - (\sigma_{\varphi}^{(n,j)})^2 \right] \end{aligned} \quad (13)$$

where $\mu_{\varphi}^{(n,j)}$ and $\sigma_{\varphi}^{(n,j)}$ denote the j th element of the mean vector $\mu_{\varphi}(\mathbf{x}_n)$ and the standard deviation vector $\sigma_{\varphi}(\mathbf{x}_n)$ of $\mathbf{z}_n$, respectively; $j = 1, \dots, J$ denotes the dimension index of the latent feature $\mathbf{z}_n$, with J representing the dimension

of $\mathbf{z}_n$. In combination with (2), (3), and (11)–(13), the eventual estimator for our mixture VAE is

$$\text{LB}(\boldsymbol{\theta}, \varphi; \mathbf{X}, \mathbf{c}) \simeq \frac{1}{I} \sum_{n=1}^N \sum_{i=1}^I \left\{ \ln p_{\theta_{c_n}}(\mathbf{x}_n | \mathbf{z}_n^i) + \frac{1}{2} \sum_{j=1}^J \left[1 + \ln(\sigma_{\varphi}^{(n,j)})^2 - (\mu_{\varphi}^{(n,j)})^2 - (\sigma_{\varphi}^{(n,j)})^2 \right] \right\}$$

with

$$\mathbf{z}_n^i = \boldsymbol{\mu}_{\varphi}(\mathbf{x}_n) + \sigma_{\varphi}(\mathbf{x}_n) \odot \boldsymbol{\varepsilon}_n^i, \boldsymbol{\varepsilon}_n^i \sim \mathcal{N}(0, \mathbf{I}) \quad (14)$$

which is analytic and can be optimized via SGA for the solutions of $\boldsymbol{\theta}$ and φ .

For the strong expression ability, the encoder and each decoder adopt the multilayer perception (MLP) architecture to realize nonlinear functions $\boldsymbol{\mu}_{\varphi}(\mathbf{x}_n)$, $\sigma_{\varphi}(\mathbf{x}_n)$, $\boldsymbol{\mu}_{\theta_{c_n}}(\mathbf{z}_n)$, and $\sigma_{\theta_{c_n}}(\mathbf{z}_n)$. An example of the MLP with single hidden layer is shown in Section F in the supplementary material.

2) *Update Component Indicator for Observations, That Is, $p(\mathbf{c}|\mathbf{X}, \boldsymbol{\theta}, \alpha)$* : A crucial step for our mixture VAE is to determine the component membership of each sample, that is, c_n . In the light of the Bayes theorem, the posterior of c_n is

$$p(c_n | \mathbf{X}, \boldsymbol{\theta}, \mathbf{c}_{-n}, \alpha) \propto p(\mathbf{X}, \boldsymbol{\theta} | \mathbf{c}, \alpha) p(c_n | \mathbf{c}_{-n}, \alpha) \propto p(c_n | \mathbf{c}_{-n}, \alpha) \prod_{n=1}^N p_{\theta_{c_n}}(\mathbf{x}_n). \quad (15)$$

Theoretically, we have

$$p(c_n | \mathbf{c}_{-n}, \alpha) = \frac{p(\mathbf{c} | \alpha)}{p(\mathbf{c}_{-n} | \alpha)} = \frac{\int p(\mathbf{c} | \boldsymbol{\pi}) p(\boldsymbol{\pi} | \alpha) d\boldsymbol{\pi}}{\int p(\mathbf{c}_{-n} | \boldsymbol{\pi}) p(\boldsymbol{\pi} | \alpha) d\boldsymbol{\pi}}. \quad (16)$$

By integrating over $\boldsymbol{\pi}$ and exploiting the property of the Dirichlet distribution [21], we can calculate the conditional probability of $c_n = k$ as $p(c_n = k | \mathbf{c}_{-n}, \alpha) = (\eta_k^{-n} + \alpha/K) / (N - 1 + \alpha)$, where η_k^{-n} represents the number of samples assigned to the k th component excluding the $\mathbf{x}_n$. When $K \rightarrow \infty$, it holds that

$$p(c_n = k | \mathbf{c}_{-n}, \alpha) = \frac{\eta_k^{-n}}{N - 1 + \alpha}. \quad (17)$$

From (15) and (17), we know that the observed sample $\mathbf{x}_n$ is assigned to the k th ($k = 1, \dots, K$) component with a prior probability $[\eta_k^{-n} / (N - 1 + \alpha)]$, and furthermore, the prior probability of $\mathbf{x}_n$ belonging to a new component is

$$p(c_n = k_{\text{new}} | \mathbf{c}_{-n}, \alpha) = 1 - \sum_{k=1}^K p(c_n = k | \mathbf{c}_{-n}, \alpha) = \frac{\alpha}{N - 1 + \alpha}. \quad (18)$$

Therefore, the indicator c_n can be confirmed via the following two criteria.

- 1) The posterior probability that a sample $\mathbf{x}_n$ attributes to component k is

$$p(c_n = k | \mathbf{X}, \boldsymbol{\theta}, \mathbf{c}_{-n}, \alpha) \propto \frac{\eta_k^{-n}}{N - 1 + \alpha} p_{\theta_k}(\mathbf{x}_n)$$

$$\approx \frac{\eta_k^{-n}}{N - 1 + \alpha} \mathbb{E}_{q_{\varphi}(\mathbf{z}_n | \mathbf{x}_n)} [\ln p_{\theta_k}(\mathbf{x}_n | \mathbf{z}_n)] \times [\ln p_{\theta_k}(\mathbf{x}_n | \mathbf{z}_n)]. \quad (19)$$

Here, we use $\mathbb{E}_{q_{\varphi}(\mathbf{z}_n | \mathbf{x}_n)} [\ln p_{\theta_k}(\mathbf{x}_n | \mathbf{z}_n)]$ to approximate $p_{\theta_k}(\mathbf{x}_n)$ since a small weight is imposed on $\text{KL}(q_{\varphi}(\mathbf{z}_n | \mathbf{x}_n) \| p_{\theta_k}(\mathbf{z}_n))$.

- 2) The probability that a sample $\mathbf{x}_n$ cannot be described by any of the current components and should be assigned to a new component is

$$p(c_n = k_{\text{new}} | \mathbf{X}, \boldsymbol{\theta}, \mathbf{c}_{-n}, \alpha) \propto \frac{\alpha}{N - 1 + \alpha} \mathbb{E}_{q_{\varphi}(\mathbf{z}_n | \mathbf{x}_n)} [\ln p_{\theta_{\text{new}}}(\mathbf{x}_n | \mathbf{z}_n)] \quad (20)$$

where θ_{new} is the initialized parameter for the new component.

As shown in (19) and (20), the posterior of parameter c_n ($n = 1, \dots, N$) given $\{\boldsymbol{\theta}, \varphi, \psi\}$ has the analytical expression based on the property of the Dirichlet distribution. Therefore, we can directly sample from (19) and (20) to obtain the estimation of c_n , rather than exploiting the variation inference, which is mainly used for the approximate solution of parameters in the inexplicit expression. In addition, from (19) and (20), we know that the DP implements the clustering and the automatic determination of the cluster number simultaneously. On the one hand, the DP formulates a clustering criterion that those observed samples with the same probability distribution are grouped together. Therefore, without DP, a new clustering criterion should be provided for our DMVAE, which is a challenge work in the context of the deep model. On the other hand, the DP also considers the probability of a sample belonging to a new cluster, and when the current clusters are insufficient to fully describe all data, DP can adjust the number of clusters according to the data distribution until makes the cluster number in the best condition, that is, the self-adaptive determination of the cluster number.

B. Discriminative Mixture Variational Autoencoder

Despite the advantage in data description, the mixture VAE is an unsupervised model that only focuses on the explanation of observations and does not utilize any supervised information. However, class labels of some samples are often accessible to many tasks, such as the classification, and it should make sense to treat the label as an available information that can be integrated into the mixture VAE. To this end, we propose a DMVAE model. The DMVAE not only models the observed data via the mixture of decoders but also depicts the generative process from the latent features to the labels of labeled data by introducing a classifier, which is utilized as the classification criteria for the proposed method. This modeling approach retains the stronger descriptive power of data and meanwhile facilitates the enhancement of discriminability in latent feature space.

Considering the whole dataset $\mathbf{X}$ as a collection of U unlabeled samples $\{\mathbf{x}_u\}_{u=1}^U$ and L labeled samples $\{\mathbf{x}_l\}_{l=1}^L$ ($U + L = N$). Let $\mathbf{y}_l$ represent the label of the l th labeled sample and be defined as a 1-of- S one-hot vector with S being the number of categories, then the DMVAE models the generation

of $\mathbf{y}_l$ as

$$p_\psi(\mathbf{y}_l|\mathbf{z}_l) = \text{Cat}(S, \lambda(\mathbf{z}_l)) \quad (21)$$

where $\lambda(\mathbf{z}_l)$ is the output of the softmax classifier with parameter ψ , $\mathbf{z}_l$ is the latent variable of $\mathbf{x}_l$ derived from the encoder, and $\text{Cat}(\cdot)$ denotes the categorical distribution. We know from (10) and (21) that the learning of latent variable is not only related to the observations but also affected by the label information. Due to the distinctions of labels from different classes, the DMVAE can obtain a more separable latent space, thus improving the classification performance.

Nevertheless, in the reality, we often have a large volume of unlabeled data, while only a small number of labeled data, that is, the semisupervised setting. In this case, the generalization capability of the classifier, trained with a few labeled samples, in our DMVAE, is poor. To tackle this problem, we introduce a regularization term, with which the predicted labels of the unlabeled data via the classifier have the minimum entropy. In detail, let $\mathbf{y}_u$ represent the label of the unlabeled sample $\mathbf{x}_u$ predicted by the classifier, its entropy is expressed as

$$H_\psi(\mathbf{y}_u) = -\mathbf{y}_u \log(\mathbf{y}_u) = -\sum_{s=1}^S y_u^{(s)} \log(y_u^{(s)}) \quad (22)$$

where $y_u^{(s)}$ is the s th ($s = 1, \dots, S$) element of $\mathbf{y}_u$. Under the accurate description of observations, the entropy constraint makes the classifier of the DMVAE possible to also have the high classification confidence for unlabeled data. This mitigates the weak prediction ability caused by the overfitting of the classifier.

In summary, the objective of the DMVAE is to get the solutions of θ , ψ , φ , and $\mathbf{c}$ by maximizing the expression of $\ln p(\mathbf{X}, \mathbf{c}, \theta|\alpha) + \beta_1 \ln p_\psi(\mathbf{Y}) - \beta_2 H_\psi(\tilde{\mathbf{Y}})$, where $\mathbf{Y}$ and $\tilde{\mathbf{Y}}$ represent the collections of labels from labeled data and predicted labels from unlabeled data, respectively; $p_\psi(\mathbf{Y}) = \prod_{l=1}^L p_\psi(\mathbf{y}_l)$ is the marginal likelihood of $\mathbf{Y}$; and β_1 and β_2 are the weights of label and entropy constraints. Similar to the approach adopted in the mixture VAE, we alternatively solve the indicator $\mathbf{c}$ and the other model parameters, including θ , ψ , and φ . For $\mathbf{c}$ given θ , ψ , and φ , it can be sampled from the posterior $p(c_n|\mathbf{X}, \theta, \mathbf{c}_{-n}, \alpha)$, whose explicit expression is presented in Section III-A; for θ , ψ , and φ given $\mathbf{c}$, they can be obtained by maximizing the following lower bound via SGA:

$$\begin{aligned} & \ln p(\mathbf{X}, \theta|\mathbf{c}) + \beta_1 \ln p_\psi(\mathbf{Y}) - \beta_2 H_\psi(\tilde{\mathbf{Y}}) \\ & \geq \text{LB}(\theta, \psi, \varphi; \mathbf{X}, \mathbf{Y}, \mathbf{c}) \\ & = \sum_{n=1}^N \left\{ \mathbb{E}_{q_\varphi(\mathbf{z}_n|\mathbf{x}_n)} [\ln p_{\theta_n}(\mathbf{x}_n|\mathbf{z}_n)] \right. \\ & \quad + \beta_1 \sum_{l=1}^L \mathbb{E}_{q_\varphi(\mathbf{z}_l|\mathbf{x}_l)} [\ln p_\psi(\mathbf{y}_l|\mathbf{z}_l)] \\ & \quad \left. - \beta_2 \sum_{u=1}^U H_\psi(\mathbf{y}_u) - \gamma \text{KL}(q_\varphi(\mathbf{z}_n|\mathbf{x}_n) \| p(\mathbf{z}_n)) \right\} \quad (23) \end{aligned}$$

and the detailed learning process is summarized in the supplementary material. It should be pointed out that the objective

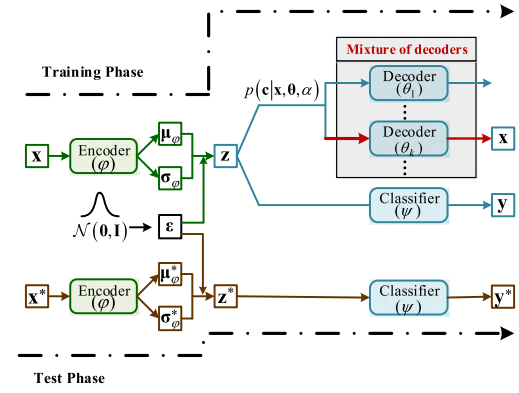


Fig. 3. Architecture of the DMVAE model. The latent variable $\mathbf{z}$ of the training sample $\mathbf{x}$, encoded via an encoder with parameter φ , is input into the specified decoder according to posterior probability $p(\mathbf{c}|\mathbf{x}, \theta, \alpha)$ and the classifier for the generations of $\mathbf{x}$ and label $\mathbf{y}$, respectively. Here, the thicker red arrows indicate more probable connection. In the test stage, the label of a test sample $\mathbf{x}^*$ can be directly obtained via the learned feedforward network.

function in the SBVAE [36] also contains the label and entropy constraints, whereas it is significantly different from (23) due to the distinction of the SBVAE and DMVAE in model structures. In detail, the SBVAE considers the sample label, also, as the latent variable and describes the observed sample as being generated by a latent feature in addition to a label variable. However, our DMVAE regards the sample label as the observed information, which is generated by the latent feature. The more detailed analysis on their objective functions is provided in Section E in the supplementary material.

Moreover, the class prediction for a test sample $\mathbf{x}^*$ can be realized by going through the feedforward propagation once the model training ends. This is more efficient than the probabilistic models [3], [4] which exploit the Markov chain Monte Carlo (MCMC) or variational Bayesian (VB) algorithm for the iterative parameter inference. For the intuitive illustration, we present the whole network architecture of the DMVAE in Fig. 3.

IV. EXPERIMENTS

In this section, two kinds of experiments are conducted to assess the performance of the proposed DMVAE model. The first experiment is the data classification task on benchmark datasets, including the MNIST, Fashion-MNIST, Heterogeneity Human Activity Recognition (HHAR) [34], and Street View House Numbers (SVHNs) [7] datasets. In the second experiment, we apply the DMVAE to radar HRRP target recognition, where the data distribution is extremely sophisticated due to the target-aspect sensitivity [1], [27], of the HRRP and enough labeled data are often not available with the limitation of radar sampling rate [27]. It should be noted that the experimental results on Fashion-MNIST, HHAR, and SVHN datasets are provided in the supplementary material due to the space limitation.

In the experiments, we initialize the cluster (decoder) number to the number of classes and randomly allocate each sample to one of the clusters. As discussed in Section III-A, our DMVAE model can self-adaptively determine the optimal



Fig. 4. Examples from each class of the MNIST dataset.

cluster number via the DP, which is validated in Section G in the supplementary material.

For the hyperparameter α in DP, it controls the prior probability that an observed sample is assigned to a cluster. To avoid the effect of an unreasonable hyperparameter setting on the clustering, we often set α to a small value and make the clustering probability, that is, the posterior probability of the indicator, only depend on the data itself. In our experiment, α is set to 1. For the weights β_1 and β_2 , they reflect the importance of the label and entropy constraints, respectively. For different datasets, the settings of β_1 and β_2 are different. Moreover, for each dataset, β_1 and β_2 should also vary with the proportion of labeled samples in all samples. Usually, when the proportion of labeled samples increases, the classifier trained with the labeled data has the stronger generalization capability and can obtain the better classification performance on the test dataset. Therefore, we should increase the value of β_1 , and meanwhile, reduce the value of β_2 to weaken the effect of entropy constraint on the accuracy of the classifier. In contrast, when the proportion of labeled samples decreases, the generalization of the classifier learned with the labeled samples is poor. In this case, β_1 should be reduced, and β_2 should be increased to improve the generalization ability of the classifier. In addition, whatever the number of labeled samples for the semisupervised DMVAE, the classifier should accurately classify the labeled samples; otherwise, the entropy constraint based on the learned classifier is unmeaningful. Consequently, the value of β_1 should be larger than that of β_2 . For γ , it reflects the influence of the KL constraint on the learning of latent features. Since the KL constraint tends to make the distributions of features from all classes close to the standard Gaussian distribution which may reduce the interclass separability of features, we often assign a small value to γ , such as 0.01 in our experiments.

A. MNIST Dataset

1) *Data Introduction and Experiment Setup*: The MNIST dataset has 60 000 training and 10 000 test samples, each of which is an image of a handwritten digit with a size of 28×28 . An example randomly selected from each class is shown in Fig. 4. The range of pixel intensity for each image is 0–255, and the normalization is implemented through dividing the pixel values by the maximum one.

In the experiments on the MNIST dataset, each image is first reshaped into a vector of dimension 784×1 and then is fed to the encoding module consisting of four fully connected layers with the numbers of hidden units in each layer being 1000, 500, 250, and 30, respectively. The symmetric architecture with the encoder, that is, 30-250-500-1000-784, is adopted for the decoder in our DMVAE. The weight initialization is conducted by exploiting the combination of fan-in and fan-out [23], and all bias parameters are initialized to zero. The

experiments are performed on a Dell PC with 3.30-GHz CPU and 8-GB RAM. Taking account of the operating efficiency and storage requirement, we select half of the training images, that is, 30 000 images, with each class having an equal number of samples to learn the DMVAE. To validate the semisupervised classification performance of our model, four different training datasets are considered: 1000/30 000, 3000/30 000, 6000/30 000, and 10 000/30 000, where 1000/30 000 means that 1000 samples are randomly selected from the 30 000 training images to form the labeled dataset and the remaining 29 000 training images are used as the unlabeled data, similarly for other expressions. In addition, the settings of β_1 under 1000, 3000, 6000, and 10 000 labeled samples are 20, 25, 30, and 30, respectively; the settings of β_2 are 5, 3, 1, and 1, respectively.

We compare our DMVAE with some related methods, including transductive SVM (TSVM) [24], VAE+TSVM, NNs [25], GMVAE [7], VaDE [34], and semisupervised VAE (SVAE). TSVM searches a reliable separating hyperplane with a transductive process that incorporates both labeled and unlabeled samples in the training phase. VAE + TSVM captures features of all samples via the traditional VAE, and then the extracted features are used to learn the TSVM classifier. NN utilizes an MLP to obtain the features of labeled data, which are simultaneously mapped to their corresponding class labels via a softmax layer. SVAE projects the latent representations of labeled samples to the labels while extracting the features with the traditional VAE (single decoder) for all observations. The same architecture as the DMVAE, that is, 784-1000-500-250-30, is designed for the above deep models.

2) *Analysis on the Training Results*: The proposed DMVAE model divides the whole dataset into several clusters and describes each cluster via a separate decoder. To yield an intuitive visualization of the clustering results, we map observations in multidimensional space to a 2-D space by using t-distribution stochastic neighbor embedding (t-SNE) and show the samples with reduced dimensionality in Fig. 5. Fig. 5(a) presents the distribution of 30 000 training samples with different colors representing different classes. The corresponding clustering results, under the semisupervised setting of 6000 labeled samples, are depicted in Fig. 5(b) where different clusters are distinguished via different colors, and the number of training samples assigned to each cluster is counted in Fig. 5(c).

Comparing Fig. 5(a) and (b), we see that the DMVAE clusters the similar samples into a group, and the whole observed space is divided into eight clusters. Taking the digit “8” as an example, its distribution has the large dispersion, and especially, there is an overlap between the digits “8” and “5,” as shown in the black box in Fig. 5(a), demonstrating the higher similarity of these samples. Via our DMVAE, the observed data mentioned above are divided into a cluster and is described with a specific decoder, which is beneficial to give a more elaborate description for these observations.

Each decoder is able to accurately represent the samples in the corresponding cluster and, thus, compared with the VAE with a single decoder (e.g., SVAE), the collection of the decoders offers the opportunity to make a more accurate

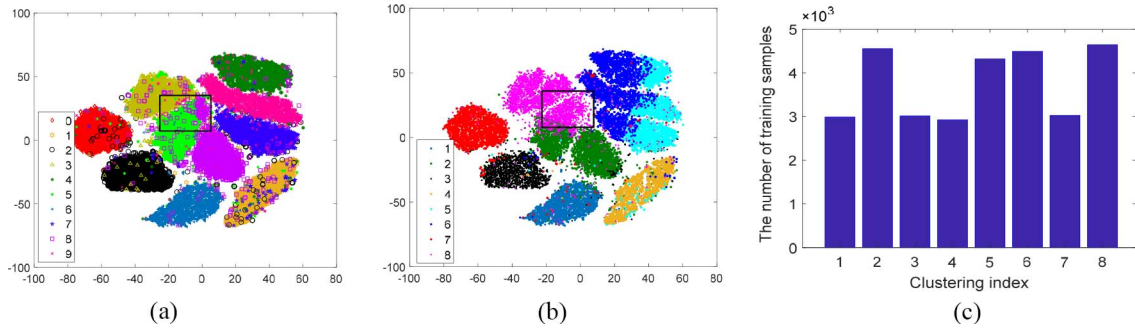


Fig. 5. Clustering visualization of training samples on the MNIST dataset with the DMVAE being trained by using 6000 labeled samples and 24 000 unlabeled samples. (a) 2-D projections of all training samples via t-SNE with each digit in the legend denoting a class. (b) Clustering results of the training samples with the digit in the legend denoting the clustering index. (c) Distribution of the training samples in each cluster. The clustering results corresponding to black boxes in (a) and (b) obviously demonstrate that DMVAE can divide the similar samples from different classes into a cluster.

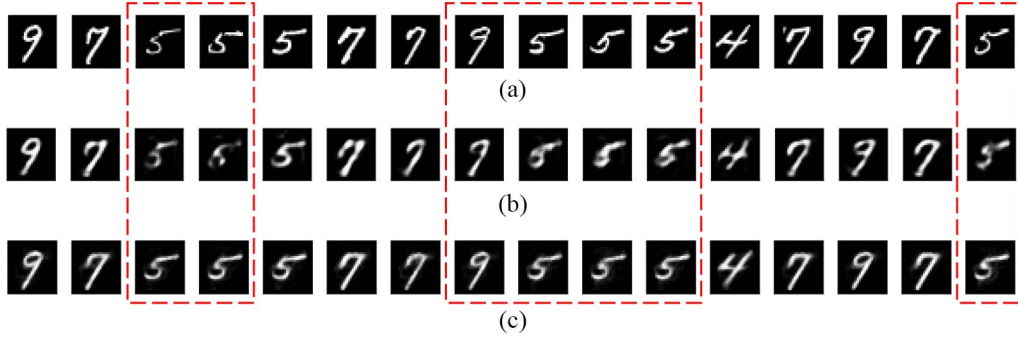


Fig. 6. Examples of the reconstructed samples via the SVAE and DMVAE models. (a) Original images. (b) Reconstructed images via the SVAE. (c) Reconstructed images via the DMVAE. Specifically, the reconstructed images framed in red dashed lines via the DMVAE are much more accurate than those via the SVAE.

depiction for the whole dataset. To demonstrate this superiority of the proposed method, we choose some samples from the training dataset and show their reconstructions via the SVAE and our DMVAE in Fig. 6. As seen from Fig. 6, our method performs better than the SVAE, especially for the images framed in red dashed lines. If we take the digit “9,” appeared in the eighth column, as an example, its reconstruction via the SVAE model looks like the digit “7,” and this poor description will result in the weak separability between the latent features of digits 9 and 7 which can be avoided in our DMVAE model due to its more precise representation for the observed data.

In Fig. 7, we quantitatively show the reconstruction errors of the training data via the SVAE and DMVAE models, under the condition of different labeled sample numbers. Here, we use the mean square error (MSE) to measure the reconstruction capability of the model. The MSE is defined as $(1/N) \sum_{n=1}^N \|\hat{\mathbf{x}}_n - \mathbf{x}_n\|_2^2$, where $\hat{\mathbf{x}}_n$ denotes the reconstructed sample corresponding to the original sample $\mathbf{x}_n$, and N denotes the number of training samples. From Fig. 7, we note that the MSEs of the two models increase with the labeled sample number, because the weight of the label constraint term should be magnified to obtain the higher classification accuracy when there are more labeled samples. This operation works at the expense of the reconstruction term, and on the contrary, limits the greater improvement of the classification performance. However, our DMVAE has the lower reconstruction errors due to the stronger capability of data description via the collection of decoders. That is, the proposed method relieves the conflict

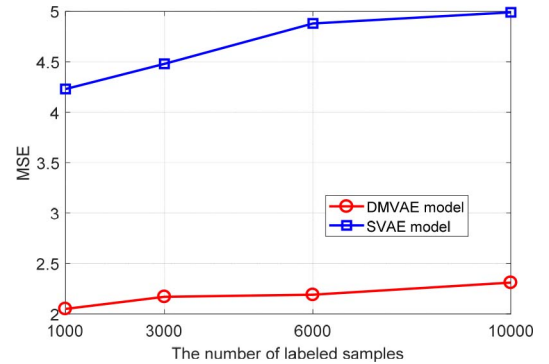


Fig. 7. Comparison of the reconstruction capability for the SVAE and DMVAE models with different numbers of labeled samples.

between the reconstruction and label constraint terms, thus can obtain the better classification performance than the SVAE.

3) *Classification on the MNIST Dataset:* In this section, we demonstrate the superiority of our DMVAE in classification performance. To give a qualitative depiction on the separability of the features, we first use t-SNE to reduce features of test samples, extracted by the GMVAE, VaDE, SVAE, and DMVAE with 6000 labeled samples, to two dimensions and then compare their separation in Fig. 8. For the MoG-VAEs, the interclass separability of features in the VaDE is superior to that in the GMVAE, due to the more reasonable modeling construction. Moreover, since our DMVAE utilizes

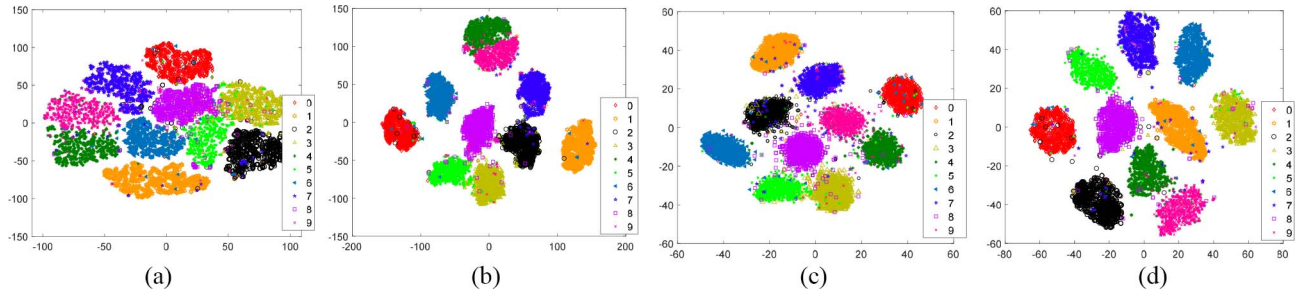


Fig. 8. 2-D t-SNE visualizations from the features of test samples extracted by the GMVAE, VaDE, SVAE, and DMVAE with 6000 labeled samples and 24 000 unlabeled samples. (a) Feature visualization via the GMVAE. (b) Feature visualization via the VaDE. (c) Feature visualization via the SVAE. (d) Feature visualization via the DMVAE.

the label information to direct feature's learning while assuring the representation ability of the feature, the DMVAE is able to learn the more separable feature space than the VaDE, as shown in Fig. 8(b) and (d). For example, the separability of digits "4" and "9" in the VaDE is obviously weaker than that in the DMVAE. In addition, since the DMVAE exploits multiple decoders to represent data more accurately, it can capture the subtle information of observations, which is beneficial to find the differences among samples, even the minor differences for those similar samples. Thus, our DMVAE is able to learn the more separable features than the SVAE model. For example, the feature separability of digits "3," "5," and "8" in our DMVAE is better than that in the SVAE, as shown in Fig. 8(c) and (d). In summary, our DMVAE has the best feature separability than the other related methods.

To quantitatively validate the effectiveness of our method, we define the within-class scatter matrix $\mathbf{S}_w$ as

$$\mathbf{S}_w = \frac{1}{N} \sum_{s=1}^S \sum_{n=1}^{N_s} (\mathbf{z}_n^s - \bar{\mathbf{z}}^s)(\mathbf{z}_n^s - \bar{\mathbf{z}}^s)^T \quad (24)$$

and the between-class scatter matrix $\mathbf{S}_b$ as

$$\mathbf{S}_b = \frac{1}{N} \sum_{s=1}^S N_s (\bar{\mathbf{z}}^s - \bar{\mathbf{z}})(\bar{\mathbf{z}}^s - \bar{\mathbf{z}})^T \quad (25)$$

where N denotes the total number of observed samples, N_s denotes the number of samples in the s th class, $\mathbf{z}_n^s$ denotes the n th feature vector from the s th class, $\bar{\mathbf{z}}^s = \sum_{n=1}^{N_s} \mathbf{z}_n^s / N_s$ is the feature mean of the s th class, and $\bar{\mathbf{z}} = \sum_{s=1}^S \sum_{n=1}^{N_s} \mathbf{z}_n^s / N$ is the mean of all feature vectors from all classes. Then, we define the ratio of $\mathbf{S}_w$ to $\mathbf{S}_b$ as $r = \text{tr}(\mathbf{S}_w) / \text{tr}(\mathbf{S}_b)$ to measure the separability of features, with $\text{tr}(\cdot)$ denoting the trace of a matrix. The smaller the value of r is, the more separable the features from different classes are. In Table II, we compare the r values of the GMVAE, VaDE, SVAE, and DMVAE, which further demonstrates the superiority of our DMVAE in separability of features.

Since the GMVAE and VaDE models can only be used for the unsupervised clustering, we mainly compare the classification performance of our DMVAE with TSVM, VAE+TSVM, NN, and SVAE methods, with their classification accuracies under different numbers of labeled samples being shown in Fig. 9. As we see from Fig. 9, the approach of VAE + TSVM has the lowest classification accuracies. This is because the

TABLE II
COMPARISON OF r VALUES WITH DIFFERENT METHODS
ON THE MNIST DATASET

Method	GMVAE	VaDE	SVAE	DMVAE
r	0.7685	0.6257	0.6302	0.5932

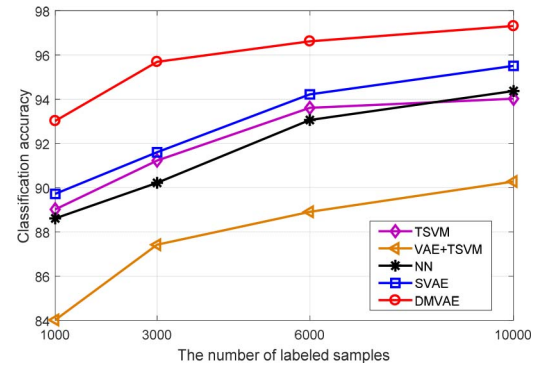


Fig. 9. Classification results of different methods under the conditions of 1000, 3000, 6000, and 10 000 labeled samples, respectively.

learning of features via the VAE and the training of the TSVM is independent, which leads to the mismatch between the feature and classifier, and moreover, the unsupervised feature extraction of VAE may also weaken the separability of observed data itself, and thus, the classification rates of the VAE+TSVM are even lower than those of the TSVM. For NN, its performance is worse than the TSVM when there are limited labeled samples, since it can only use the labeled data to learn the network. However, when the labeled samples increase to some extent, the correctness in the classification of the NN is higher than that of the TSVM due to the joint learning of features and classifier. Both the SVAE and our DMVAE have the better performance than the TSVM under a few labeled samples because of features' stronger discrimination via the label constraint. In addition, our DMVAE integrates clustering with the traditional VAE, which has the capacity to make a more accurate description for observations, thus further improving the classification ability of the model.

As demonstrated in Section III-B, besides the labeled data, the minimum entropy constraint is also helpful for the learning

TABLE III
CLASSIFICATION RESULTS (%) OF THE DMVAE WITH AND WITHOUT
MINIMUM ENTROPY CONSTRAINT

Method/Labels	1000	3000	6000	10000
DMVAE/O	90.45	94.67	96.38	97.10
DMVAE/W	93.02	95.69	96.62	97.31

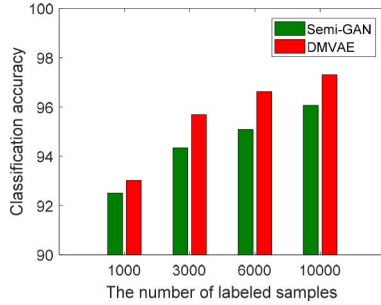


Fig. 10. Classification accuracies of the DMVAE and semi-GAN on the MNIST dataset under the different labeled sample number conditions.

of classifier. In Table III, we present the classification accuracies of DMVAE with (DMVAE/W) and without (DMVAE/O) entropy constraint. If we only use labeled samples to train the classifier, it will encounter the problem of bad generalization when the labeled samples are insufficient. By introducing the minimum entropy constraint of unlabeled data, the interplay of precise data description, label, and entropy constraints can not only guarantee the accurate classification of labeled samples but also strengthen the classification confidence of unlabeled samples, thus ultimately improving the generalization ability of the classifier, especially when there are only a few labeled samples, as shown in Table III.

4) *Performance Comparison of DMVAE and GAN-Based Models:* Besides the VAE-based models, the generative adversarial network (GAN) is also a kind of the deep generative model. GAN contains a generator and a discriminator. The generator aims at translating the noise samples into the fake samples with high similarity to the true observed data. The discriminator takes the fake and true data as the inputs and outputs their corresponding labels to distinguish the fake samples from true samples. Through the alternative iteration for the generator and discriminator, the GAN can ultimately generate the new data similar to the true observations via the generator.

Due to the generation ability, the GAN-based models are used in many applications, as the data augmentation and data transformation between two different domains. In particular, Salimans *et al.* [41] proposed a semisupervised classification method (called semi-GAN for convenience), which utilizes the generated data produced by GAN. To compare the performance of our DMVAE and GAN-based methods, Fig. 10 presents the classification accuracies of the semi-GAN and DMVAE on the MNIST dataset, under different labeled sample number conditions, from which we note that our DMVAE is better than the semi-GAN. This is because the quality of the generated samples has a significant effect on

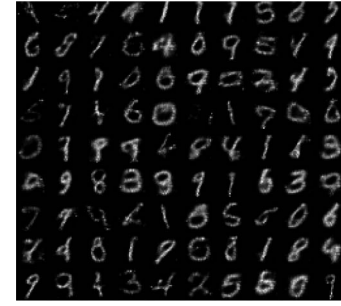


Fig. 11. Examples of generated data via the learned semi-GAN.

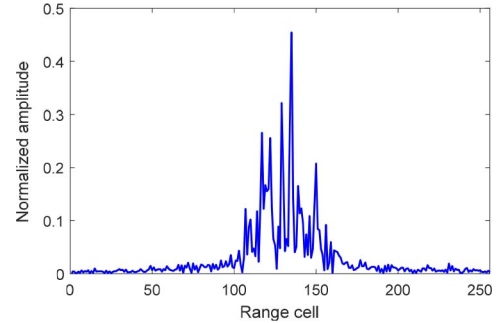


Fig. 12. Example of the HRRP waveform.

the feature learning, although the generated data play a role of data augmentation in the model learning. The good generated data will promote the feature learning toward the direction of more separable, and the bad generated data will weaken the feature separability. In Fig. 11, we give some examples of generated images via the semi-GAN on the MNIST dataset, from which we see that the semi-GAN has the poor generative capability, thus leading to the worse classification performance than our DMVAE.

B. Measured Radar HRRP Dataset

1) *Data Description:* The radar target can be described with the scattering center model [26]. This model considers the target echo along with the direction of radar light of sight (LOS) as the coherent summations of complex time returns from target scatterers in each range resolution cell, whose amplitude is defined as the radar HRRP. An example of the HRRP sample is presented in Fig. 12. Due to the reflection of target structures, such as target shape, size, and so on, the topic of HRRP-based radar target recognition is widely attractive [1], [26]–[29]. However, the HRRP is very sensitive to the change of target altitude, and HRRPs from different altitudes will have great differences when there is a considerable target-altitude variation, that is, target-aspect sensitivity of HRRP [1], [27], which leads to observations' extremely complicated distribution. Therefore, it is a hotspot that how to accurately describe the distribution characteristics of HRRPs with an appropriate parametric model, and the proposal of our DMVAE is of practical significance.

In the experiment, the measured HRRP data recorded by radar are from three real airplanes, including Yark-42,

TABLE IV
PARAMETERS OF THREE AIRPLANES AND RADAR IN THE EXPERIMENT

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	height (m)
Yark-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

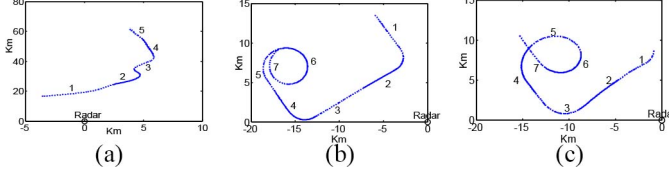


Fig. 13. 2-D track charts of three radar targets. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

Cessna Citation S/II, and An-26. Table IV presents the airplanes' sizes and radar measurement parameters, and the 2-D track charts of three targets are shown in Fig. 13, from which we see that each target track is split into several segments. In each segment, there are 25 600 HRRP samples with each sample of size 256×1 . We select 7000 HRRP samples from the 2nd and 5th segments of Yark-42, 10 000 HRRPs from the 6th and 7th segments of Cessna Citation S/II, and 10 000 HRRPs from the 5th and 6th segments of An-26 as the training samples, and 5200 HRRPs from other data segments of three targets as the test samples. In addition, 540, 1350, 2700, and 13 500 HRRPs randomly selected from the whole training dataset are used as the labeled samples, respectively, to analyze the semisupervised recognition performance of the DMVAE. The settings of β_1 under 540, 1350, 2700, and 13 500 labeled samples are 20, 25, 25, and 30, respectively; the settings of β_2 are 1, 1, 0.8, and 0.5, respectively. Moreover, before applying DMVAE to HRRP, sliding correlation processing [1] and L_2 normalization [1], [27] techniques are adopted to deal with the time-shift and amplitude-scale sensitivities of HRRP.

In this section, besides comparative methods mentioned in Section IV-A, three commonly used models for HRRP-based target recognition, including two shallow methods, that is, K-SVD [28] and FA [27], and a deep model called stacked corrective autoencoder (SCAE) [29], are also added to fully demonstrate the superiority of the proposed method.

2) *Performance Analysis on the HRRP Dataset*: In Fig. 14, we visualize the training samples' clustering results via our DMVAE model under the semisupervised setting of 540 labeled samples. Note from Fig. 14(a) that the HRRP dataset is complicatedly distributed. Specifically, the targets of Yark-42 and An-26 present the characteristic of multimode distribution. Therefore, it is hard to give a fine and precise descriptions of observations with a single model. However, our DMVAE model partitions the whole training dataset into five clusters, and the targets with multiple modes are clustered into multiple groups, as shown in Fig. 14(b). Since each cluster

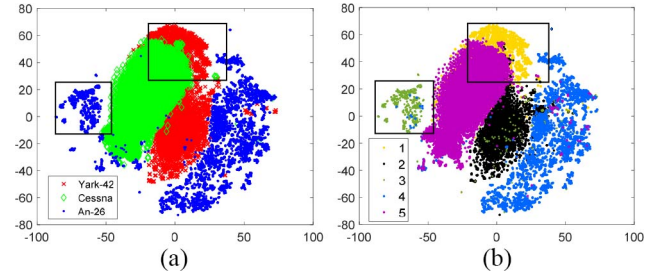


Fig. 14. Clustering visualization of HRRP training data with the DMVAE being trained by using 540 labeled samples and 26 460 unlabeled samples. (a) 2-D projections of all training samples via t-SNE. (b) Clustering results of the training samples with the digit in the legend denoting the clustering index. The clustering results in the black box explicitly demonstrate that DMVAE can divide the similar samples into the same cluster and samples with larger differences into the different clusters, even though they are from the same class.

TABLE V
RECONSTRUCTION ERROR (MSE) OF THE HRRP TRAINING DATASET VIA THE SVAE AND DMVAE MODELS

Method/Labels	540	1350	2700	13500
SVAE	0.3520	0.3537	0.3560	0.3577
DMVAE	0.2676	0.2834	0.2968	0.3034

is described through a specific decoder which can accurately represent observations with simple distribution, the collection of decoders in our DMVAE can achieve better descriptive performance than the model with a single decoder.

To illustrate the superiority of our DMVAE in description capability, we randomly select three samples from the training HRRP dataset and show their reconstructions via the SVAE and DMVAE, respectively, in Fig. 15. From Fig. 15, we note that the basic shapes of reconstructed HRRPs via two models are consistent with those of the original samples, however, the DMVAE has the better description for the subtle structures of HRRPs than the SVAE, such as the more accurate reconstruction of the peak values of HRRPs. Table V quantitatively compares the reconstruction error of training samples with above two models, further demonstrating the preponderance of the proposed method.

To validate the advantage of the proposed method in feature separability, we project the latent features of test samples, learned by the GMVAE, VaDE, SVAE, and DMVAE models with 540 labeled samples and 26 460 unlabeled samples, to the 2-D space via the t-SNE, and compare their separability in Fig. 16. For the features of Yark-42 and features from the other two targets learned with four models, there is a clear distinction, but the DMVAE method has the greater interclass distances than the other three methods. For the targets of Cessna Citation S/II and An-26, their features learned by GMVAE, VaDE, and SVAE models have the larger overlaps than those learned by the DMVAE. In summary, our DMVAE has the better feature separability, which can be quantitatively validated by the ratio of within-class scatter matrix to between-class scatter matrix shown in Table VI.

Comparisons of recognition performances under different numbers of labeled samples, via the DMVAE and other related

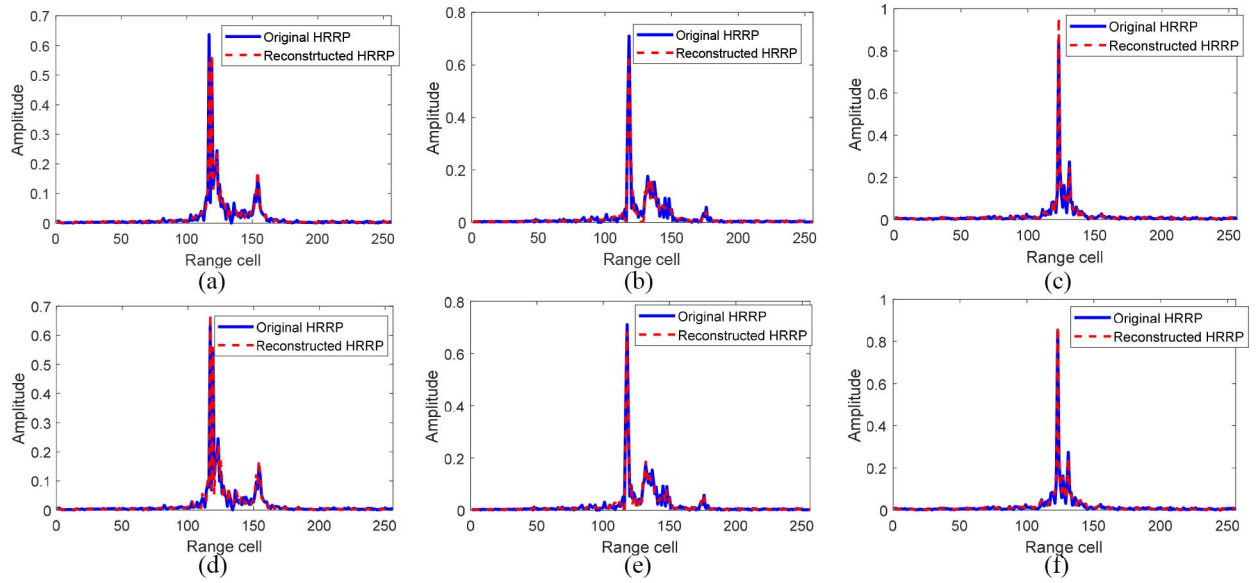


Fig. 15. Reconstructions of HRRP samples via the SVAE and DMVAE models. (a), (c), and (e) Examples of reconstructed HRRPs via the SVAE model. (b), (d), and (f) Examples of reconstructed HRRPs via the DMVAE model.

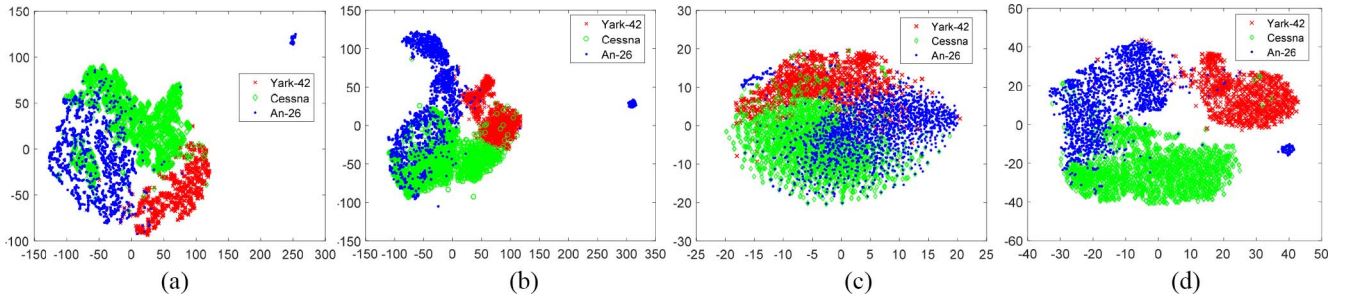


Fig. 16. 2-D t-SNE visualizations from the features of test samples extracted by the GMVAE, VaDE, SVAE, and DMVAE with 540 labeled samples and 26 460 unlabeled samples. (a) Feature visualization via the GMVAE. (b) Feature visualization via the VaDE. (c) Feature visualization via the SVAE. (d) Feature visualization via the DMVAE.

TABLE VI
COMPARISON OF r VALUES WITH DIFFERENT METHODS
ON THE HRRP DATASET

Method	GMVAE	VaDE	SVAE	DMVAE
r	2.2035	1.9806	2.2157	1.0214

models, are depicted in Fig. 17. The shallow models, including K-SVD and FA, are supervised and are sensitive to the number of training samples due to their higher model complexity and, thus, the recognition rates of these two models are lower than those of the TSVM and dramatically decrease with the reduction of labeled samples. Because of the weakness of separability caused by unsupervised feature extraction, the recognition performance of VAE+TSVM is worse than that of the TSVM. NN outperforms than the TSVM in the case of enough labeled samples, however, its performance becomes bad when there are only a few labeled samples. For SCAE, its recognition ability is superior to that of the TSVM due to the introduction of the corrective term. The SVAE adopts the mechanism of semisupervised learning and thus obtains

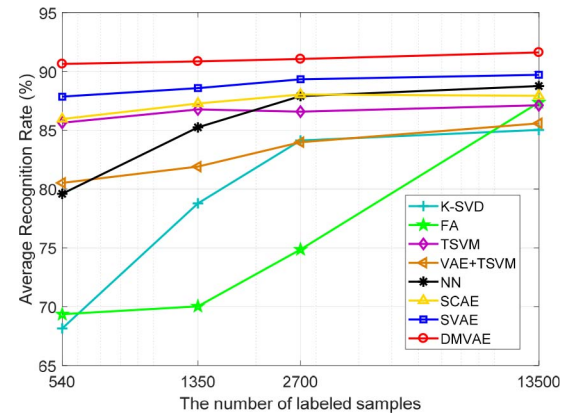


Fig. 17. Recognition performances of the DMVAE and other related models with different numbers of labeled samples on the HRRP dataset.

the superior performance than the NN, especially when there are only a few labeled samples. Furthermore, DMVAE pulls a more accurate description of observations, classifier learning, and entropy constraints together, thus achieving the best recognition capability compared with other related models.

V. CONCLUSION

A semisupervised classification method via the deep probability model is discussed in this article. To achieve a more accurate description of observations, we propose a probabilistic autoencoder called DMVAE. In the proposed method, the whole observed dataset is automatically divided into several clusters via the DP prior and the observations in each cluster are described with a specific decoder. Due to the stronger description capability of the single decoder for a small patch of the observed space, the mixtures of multiple decoders can give a more precise depiction of all observed data, which avoids the poor representation of features caused by inaccurate description to observations. In addition, the label information from the labeled samples is exploited via introducing a classifier, with which the minimum entropy can be guaranteed. With the accurate representation of observations, label, and entropy constraints, not only the features from different classes are more separable but also the generalization performance of the classifier in the DMVAE is greatly enhanced, ultimately helpful for the achievement of the higher classification accuracy. Experiments on benchmark datasets and measured radar HRRP data validate the effectiveness of our DMVAE model.

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